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COMPARISON OF THE POTENCIES OF CONGENERS OF LSD IN INDUCING PSYCHOSES WITH POTENCIES IN BLOCKING SEROTONIN INDUCED CONTRACTION IN SMOOTH MUSCLE

DRUG	ANTISEROTONIN POTENCY[1]	PSYCHOTOMIMETIC POTENCY[2]
d-DIETHYLAMIDE OF LYSERGIC ACID (LSD)	100	100
A. ISOMERS		
L-DIETHYLAMIDE (L-LSD)	0.06	<0.025
d-ISO-DIETHYLAMIDE (ILD)	0.12	<2.0
B. DISUBSTITUTED AMIDES		
d-LYSERGIC ACID AMIDE (LA III)	4	<3.0
d-LYSERGIC ACID DIETHYLAMIDE (DAM)	23	13
C. MONOSUBSTITUTED AMIDES		
d-MONOMETHYL AMIDE	6	<5
d-MONOETHYL AMIDE (LAE)	12	5
D. CYCLIC AMIDES		
d-MORPHOLIDE (LSM)	2.0	10
d-PYROLLIDID (LPD)	5.0	6
E. REMOVAL OF ACID AMIDE		
d-LYSERGIC ACID	<1	<3.0
F. RING-SUBSTITUTED DERIVATIVES		
(1) ON INDOLE NITROGEN		
1-ACETYL LSD (ALD)	210	>100
1-METHYL LSD (MLD)	368	33
(2) ON CARBON 2		
2-BROM-LSD (BOL)	103	<0.5
(3) ON INDOLE NITROGEN AND CARBON 2		
1-METHYL-2-BROM LSD (MBL)	533	<0.5
(4) ON INDOLE NITROGEN AND VARIATION IN ACID AMIDE		
1-METHYL-MONOETHYL AMIDE (MLA)	11	<4
1-ACETYL-MONOETHYL AMIDE (ALA)	11	<6
1-METHYL-PYROLLADID (MPD)	5	<4

1. FROM DATA OF CERLETTI AND DOEPFNER, J.P.E.T. 122:124-136 (JAN.) 1958
2. EXPRESSED AS RATIO ; $\frac{\text{POTENCY OF LSD (1 MCG/KG)}}{\text{POTENCY OF CONGENER}} \times 100$

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